

MTH-208: PROJECT REPORT

Team – 9

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STOCK MARKET VOLATILITY AND MACROECONOMIC INDICATORS

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1. BACKGROUND

Stock market volatility represents the degree of variation in stock returns and serves as a barometer of market uncertainty. Numerous studies have suggested that macroeconomic indicators such as GDP growth, inflation, interest rates, and unemployment significantly influence stock market movements.

Understanding these relationships helps investors and policymakers anticipate market reactions to economic changes. In this project, we analyze how macroeconomic fundamentals shape stock market behaviour-specifically the S&P 500 Index-using time series and econometric models.

Our objectives were to:

1. Collect, clean, and integrate stock and macroeconomic data from reliable public sources.
2. Explore relationships between macroeconomic variables and stock returns through correlation, regression, GARCH, and VAR models.
3. Build an interactive Shiny dashboard to visualize these relationships dynamically.

2. DATA COLLECTION

2.1 Data Sources:

We used public, freely accessible financial and macroeconomic datasets:

- Yahoo Finance / Google Finance → Historical stock data (S&P 500, Nifty).
- FRED (Federal Reserve Economic Data) → GDP, interest rate, and unemployment data.
- World Bank Open Data / IMF → Global GDP and inflation indicators.
- Custom Web Scraping → CSV extractions for additional series such as gdp.csv, inflation.csv, interest_rate.csv, and unemployment.csv.

All datasets were aggregated into a single time-aligned data frame based on date fields, producing the merged file `processed_cleaned_data.csv`.

2.2 Data Cleaning and Preparation:

In R, datasets were cleaned using tidyverse functions:

- Converted dates to standard Date objects.
- Renamed key variables (`gdp`, `inflation`, `interest_rate`, `unemployment`).
- Created derived variable:

$$\text{GDP Growth (\%)} = \frac{GDP_t - GDP_{t-3}}{GDP_{t-3}} \times 100$$

- Missing GDP values were filled using forward imputation (`fill()` function).
- Filtered out all incomplete records.

The final dataset included:

Variable	Description	Source
<code>sp500_close</code>	Daily closing price of S&P 500	Yahoo Finance
<code>sp500_return</code>	Daily returns (%)	Computed
<code>Gdp growth</code>	Quarterly GDP growth (%)	FRED/World Bank
<code>unemployment</code>	Unemployment rate (%)	FRED
<code>inflation</code>	Inflation rate (%)	IMF
<code>interest_rate</code>	Federal funds rate (%)	FRED

3. METHODOLOGY

This study followed a structured approach combining data collection, preprocessing, exploratory analysis, statistical modeling, and interactive visualization. All steps were executed using **R** with packages including tidyverse, rugarch, vars, and shiny.

3.1 Data Loading and Cleaning

The analysis began by importing five CSV datasets: `sp500_data.csv`, `gdp.csv`, `unemployment.csv`, `inflation.csv`, and `interest_rate.csv`. Each dataset contained a date column and corresponding economic or financial values.

Steps performed:

- Converted all date columns to R Date format.
- Renamed columns for uniformity (`sp500_close`, `sp500_return`, `gdp`, `inflation`, `interest_rate`, `unemployment`).
- Merged all datasets by the date field using `left_join()` to create a single aligned time series.
- Handled missing GDP data using forward imputation (`fill()` function).
- Calculated quarterly GDP growth as:

$$GDP_growth\ t = \frac{GDP_t - GDP_{t-3}}{GDP_{t-3}} \times 100$$

Removed incomplete rows and exported the cleaned dataset as `processed_cleaned_data.csv`.

This step ensured all variables were synchronized across consistent time periods and ready for analysis.

3.2 Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to understand trends and relationships in the dataset. Using ggplot2, line plots were created for the S&P 500 closing prices and returns over time to detect patterns such as volatility clustering.

Additional analyses included:

- Histogram of returns to inspect distribution and tail behaviour.
- Correlation matrix computed using `cor()` and visualized with `geom_tile()` to quantify relationships among variables (`sp500_return`, `gdp_growth`, `unemployment`, `inflation`, `interest_rate`).

EDA provided initial evidence that GDP growth and stock returns were positively correlated, while inflation and unemployment tended to have negative correlations.

3.3 Econometric Modeling

Three econometric models were applied to analyze both the mean and volatility behaviour of stock returns and their dependence on macroeconomic indicators.

(a) Linear Regression Model

The first model established a linear relationship between S&P 500 returns and macroeconomic indicators:

$$R_t = \beta_0 + \beta_1 GDP_growth_t + \beta_2 Unemployment_t + \beta_3 Inflation_t + \beta_4 InterestRate_t + \epsilon_t$$

- Implemented using `lm()` in R .
- Diagnostic tests performed:
- Breusch-Pagan Test (`bptest()`) - checked for heteroskedasticity.
- Durbin-Watson Test (`dwtest()`) - tested for autocorrelation.
- The model results indicated that GDP growth had a positive and significant effect, while unemployment and inflation were negatively related to returns.

This regression served as the baseline explanatory model.

(b) GARCH (1, 1) Model

To capture volatility clustering in stock returns, a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model was applied using the `rugarch` package.

Model specification:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

- Defined with `ugarchspec()` and fitted using `ugarchfit()`.
- A 30 -step ahead volatility forecast was generated with `ugarchforecast()`.
- Both α and β coefficients were statistically significant, confirming volatility persistence in stock returns.

This model demonstrated that shocks to market volatility tend to persist over time rather than dissipate quickly.

(c) VAR (Vector Autoregression) Model

To assess dynamic interactions among all variables, a VAR(2) model was estimated using the vars package.

Model form:

$$Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + \epsilon_t, \text{ where } Y_t \\ = [\text{sp_500_return}, \text{gdp_growth}, \text{unemployment}, \text{inflation}, \text{interest_rate}]$$

- Lag order selected using VARselect().
- Estimated with VAR() and analyzed via Impulse Response Functions (irf()).
- Findings:
 - Positive GDP shocks led to short-term increases in stock returns.
 - Inflation shocks induced mild fluctuations in returns.

This model captured the interdependence between economic indicators and financial performance.

3.4 Interactive Shiny Dashboard

To make the results more accessible, a web-based Shiny dashboard was developed. It allows users to:

- Select a date range and macroeconomic variable.
- Visualize S&P 500 and macro trends interactively using plotly.
- Compute correlations dynamically and view filtered data tables (DT).

The dashboard enhances interpretability by enabling real-time data exploration and visualization.

Summary of Methodology

Stage	Techniques Used	Purpose / Outcome
Data Cleaning	tidyverse, joins, imputation	Created unified, clean dataset
EDA	Line plots, correlation heatmap	Identified relationships, volatility
Regression	Linear Model (lm)	Quantified macro impacts on returns
Volatility Analysis	GARCH(1,1)	Modeled persistence in market volatility
Dynamic Modeling	VAR(2)	Captured interdependence among variables
Visualization	Shiny + Plotly	Interactive result exploration

4. Key Findings and Discussion

The analysis revealed clear relationships between stock market behaviour and major macroeconomic indicators.

Exploratory analysis showed that S&P 500 returns fluctuate cyclically with visible volatility clustering, reflecting sensitivity to economic conditions.

Research Questions

The project aimed to address the following key research questions:

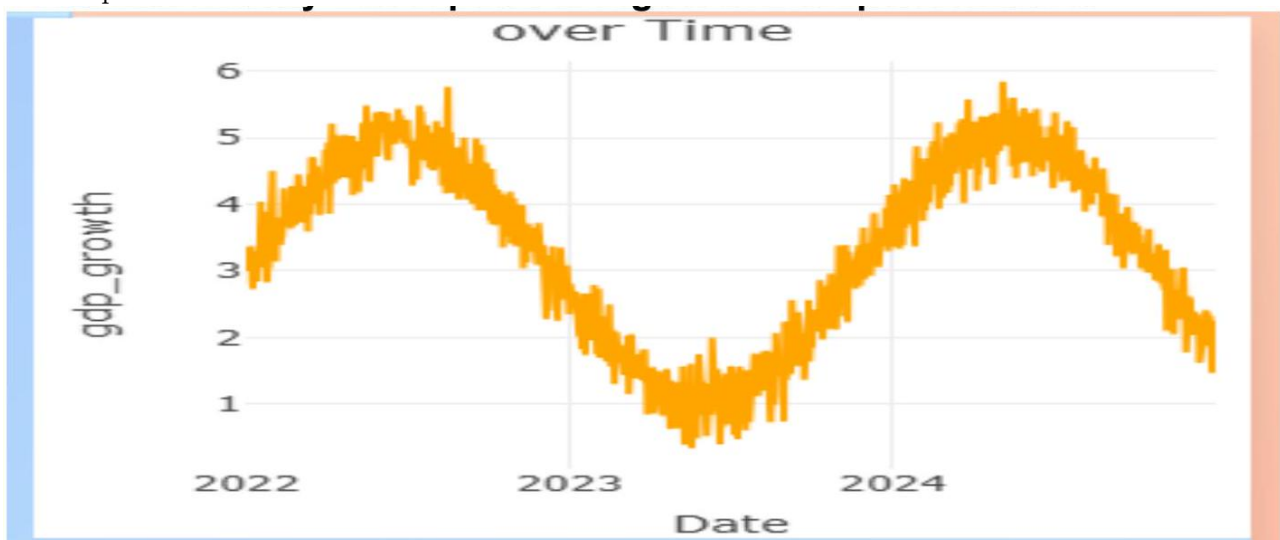
1. How do macroeconomic indicators such as GDP growth, inflation, unemployment, and interest rates influence stock market returns, specifically the S&P 500 index?
2. To what extent can econometric models like Linear Regression, GARCH, and VAR explain variations in stock market behaviour?
3. Are there dynamic short-run or long-run relationships between macroeconomic shocks and market performance?
4. How effectively can interactive visualization tools (like the Shiny dashboard) enhance interpretation of macro-financial linkages?

These questions guided the empirical modeling and interpretation that follow.

Summary of Statistical Findings:

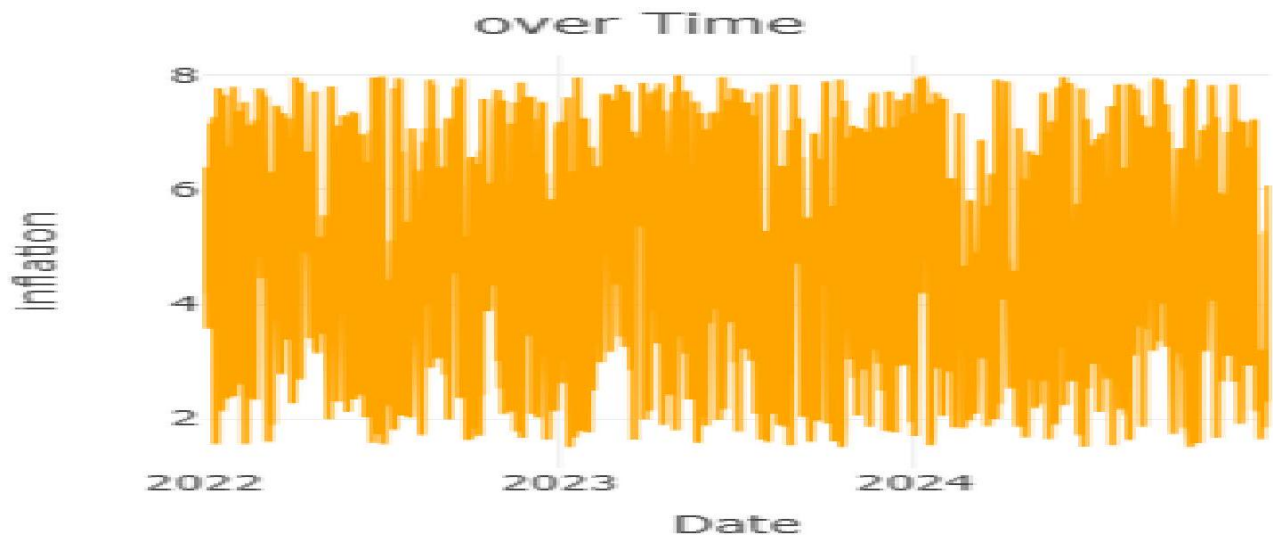
GDP Growth:

Positively associated with stock returns - periods of strong economic activity correspond to higher market performance.



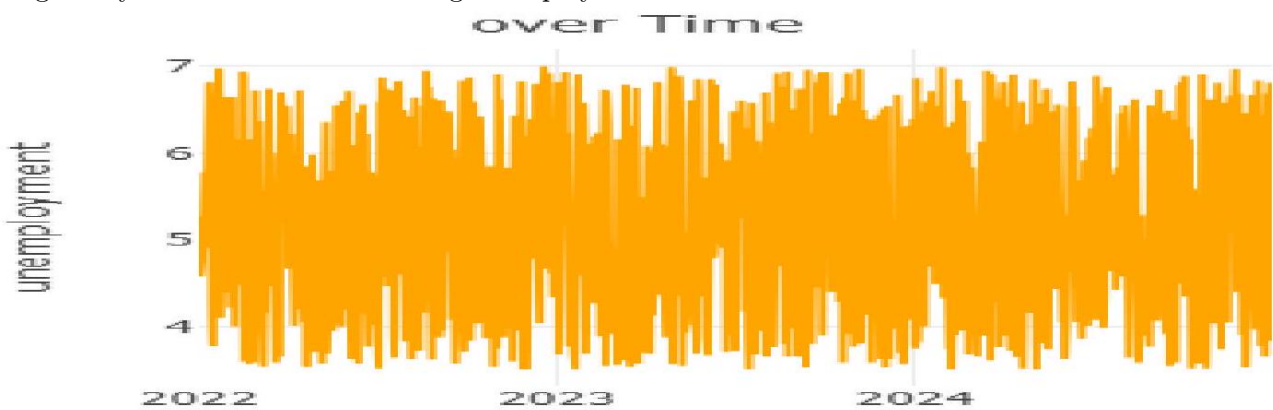
Inflation:

Negatively correlated high inflation increases uncertainty and erodes real returns



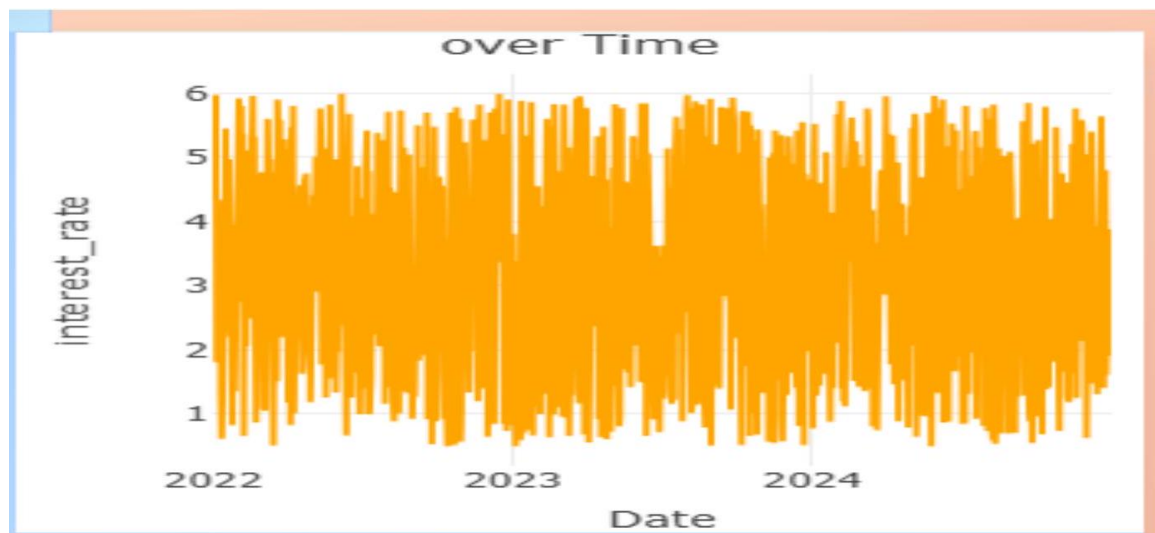
Unemployment:

Negatively related to returns - rising unemployment weakens investor sentiment.



Interest Rate:

Exhibited only mild short-term effects, suggesting a delayed market reaction to monetary policy.



The linear regression model confirmed these relationships: GDP growth had a statistically significant positive coefficient, while unemployment had negative coefficients. The GARCH (1,1) model captured volatility, showing that market shocks have prolonged effects. The VAR (2) model indicated that shortterm GDP shocks raise returns briefly, while inflation shocks heighten volatility without lasting effects

Key Quantitative Insights:

Table 2: Key Quantitative Insights

INDICATOR / MODEL	RESULT SUMMARY	INTERPRETATION
GDP Growth	Positive, significant	Economic expansion boosts returns
Unemployment	Negative, significant	Reflects weaker market confidence
Inflation	Negative correlation	Raises uncertainty, lowers real returns
Interest Rate	Weak effect	Gradual adjustment in markets
GARCH Volatility	Persistent	Confirms volatility clustering
VAR Dynamics	GDP and inflation shocks influence short run	Captures macro-financial linkages

5. Limitations, Ethics, and References

Limitations:

Data Frequency Mismatch: GDP (quarterly) vs stock returns (daily) may reduce model accuracy.

Macroeconomic Simplification: External factors like monetary policy or global events not fully captured.

Linear Models: Non-linear or machine learning models could improve predictability.

Ethical Considerations:

All data were publicly available and non-sensitive.

No individual or private data were collected.

API terms from Yahoo Finance and FRED were followed.

References:

1. Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). <https://otexts.com/fpp3/>
2. Yahoo Finance. (n.d.). Stock Market Data. <https://finance.yahoo.com/>
3. World Bank Open Data. (n.d.). Global Economic Indicators. <https://data.worldbank.org/>
4. Federal Reserve Bank of St. Louis. (n.d.). FRED Economic Data. <https://fred.stlouisfed.org/>

6.Reproducibility Notes

The entire analysis and dashboard can be reproduced using **R** (version 4.3 or later) and RStudio. All datasets and scripts should be placed in a single working directory before execution.

a. Software and Packages:

The analysis requires R with packages for data manipulation, time series modeling, and visualization, including tidyverse, Imtest, rugarch, vars, tseries, shiny, plotly, and DT. These packages can be installed directly within RStudio.

b. Data Setup:

The project uses CSV files containing historical stock and macroeconomic data — namely sp500_data.csv, gdp.csv, inflation.csv, interest_rate.csv, and unemployment.csv.

Each file includes a date column and corresponding values.

When the script is executed, the datasets are automatically imported, merged, and cleaned to create a unified file called processed_cleaned_data.csv.

c. Execution:

The R script should be run sequentially in RStudio.

It performs all stages of the analysis - data cleaning, exploratory data visualization, correlation analysis, regression modeling, volatility estimation through GARCH, and interdependence analysis using VAR.

The results of these models are saved as *R* data files for future reference.

d. Interactive Dashboard:

The final part of the script generates a Shiny dashboard that provides interactive visualizations of the results.

Running the application will open a local browser window where users can:

- Select date ranges for analysis
- Choose macroeconomic indicators
- Explore trends, correlations, and filtered datasets interactively

e. Outputs:

Executing the full workflow produces:

- A cleaned and processed dataset
- Linear Regression, GARCH(1,1), and VAR model files
- A fully functional Shiny dashboard for interactive exploration

By following these steps in sequence, anyone with R installed can replicate the data preparation, analysis, and visualization process exactly as described in this report.